Code No: 156AH **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) What are the advantages of a compiler over an interpreter?
b) Draw the structure of a compiler and describe various phases in the compilation process mention the output of the following statement: $id_1=id_2+id_3*50$ at each phase. [5+10]

2.a) Design the LEX program that recognizes the tokens of a C language and returns the token found.
b) Give the DFA and NFA to accept the strings containing a, b such that the string contains even number of a's and odd number of b's. [7+8]

3.a) Remove the left recursion for the following grammar and also find FIRSTS and FOLLOWs.
E→E + T | T
T→T * F | F
F → (E)/id
b) Write the steps/algorithm to construct the predictive parser table and explain with an example. [7+8]

4.a) Construct the Recursive Descent Parser with backtracking for the following grammar:
S→aSbS | bSaS | Ɛ
b) Compute LR(0) items for the following grammar and construct SLR parser table: [7+8]
S→L=R | R
L→ *R | id
R→L

5.a) Construct the syntax directed definition to convert infix notation into postfix notation.
b) Describe different ways of implementing intermediate code generation of a three-address statement. [8+7]

6.a) Explain how an L-attributed grammar is converted into a translation scheme.
b) Compare and contrast S-Attributed definitions with L-Attributed definitions. [8+7]

7. How is stack storage allocation strategy different from heap allocation strategy? Describe them mentioning their merits and demerits. [15]

8. Explain the foundations and basic notations used in data-flow analysis for optimizations with examples. [15]

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